

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

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Coursera - IBM Data Science Professional Certificate

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Project Background:

- Commercial space company SpaceX advertises Falcon 9 rocket launches at \$62 million, significantly cheaper than competitors.
- The cost savings are primarily due to SpaceX's ability to reuse the first stage of the rocket.

Problem Statement:

- Our goal is to predict whether the Falcon 9 first stage will land successfully.
- Accurately predicting this outcome helps estimate launch costs and provides valuable insights for competing companies.

GitHub Repository URL:

<https://github.com/huyennt92/IBM-Data-Science-Capstone>

Executive Summary

Methodology:

- **Data Collection:** Utilized SpaceX REST API and Web Scraping from Wikipedia.
- **Data Wrangling:** Cleaned data, handled missing values, and applied One-Hot Encoding using Pandas.
- **EDA:** Conducted Exploratory Data Analysis using SQL (DB2) and Visualization (Matplotlib/Seaborn).
- **Interactive Analytics:** Built interactive maps with Folium and dashboards with Plotly Dash.
- **Predictive Analysis:** Trained Logistic Regression, SVM, Decision Tree, and KNN models.

Key Results:

- The Decision Tree model achieved the highest prediction accuracy on the test data.
- Payload mass and launch site are significant indicators of landing success.

Process Overview:

- Requested rocket launch data from the SpaceX REST API using `requests.get()`.
- Converted the JSON response into a Pandas DataFrame using `json_normalize`.
- Filtered data to include only Falcon 9 launches.
- Replaced missing Payload Mass values with the calculated mean.

GitHub URL for API Notebook:

<https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Process Overview:

- Requested historical Falcon 9 launch records from Wikipedia.
- Parsed the HTML response using the BeautifulSoup library.
- Extracted column names from HTML table headers (<th>).
- Iterated through table rows (<tr>) and cells (<td>) to construct a structured Pandas DataFrame.

GitHub URL for Web Scraping Notebook:

<https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>

Cleaning and Processing:

- Calculated the number of successful and failed landings based on the Outcome column.
- Created a new binary target variable Class (1 = Success, 0 = Failure).
- Applied **One-Hot Encoding** using `pd.get_dummies()` on categorical variables such as Orbit, LaunchSite, LandingPad, and Serial.
- Cast all numerical features to `float64` to prepare for Machine Learning models.

GitHub URL for Data Wrangling Notebook:

<https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

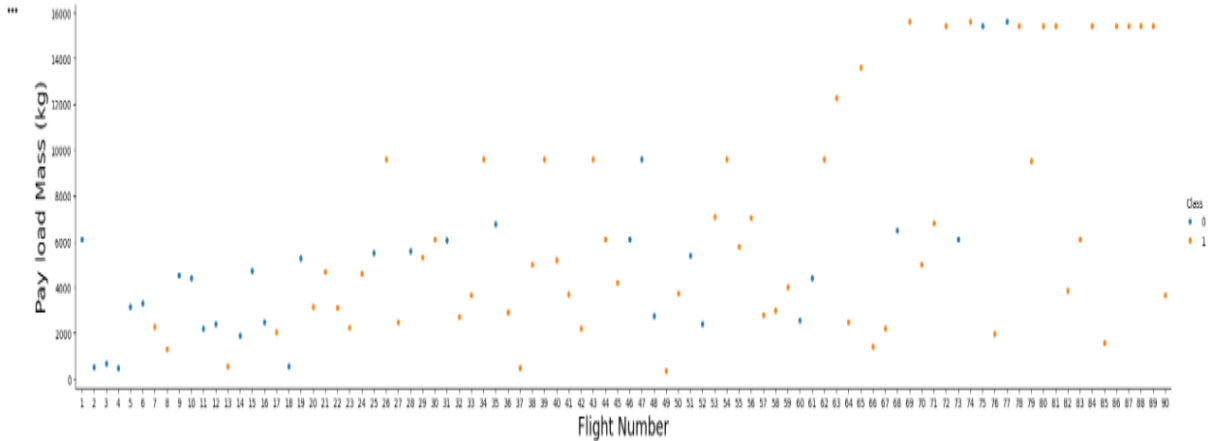
Charts Used and Their Purpose:

- **Scatter Plots:** Used to visualize the relationship between *Flight Number* vs. *Launch Site* and *Payload Mass* vs. *Launch Site*, colored by landing class.
- **Bar Charts:** Used to display the success rate across different *Orbit* types.
- **Line Charts:** Used to track the *success rate over the years* (annual trends).

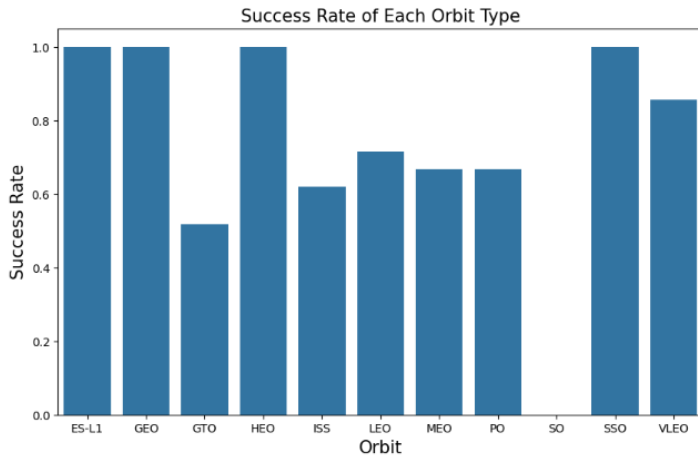
GitHub URL for EDA Visualization Notebook:

https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA Visualization Results 1



EDA Visualization Results 2



SQL Queries Performed:

- ✓ Queried unique Launch Sites from the database.
- ✓ Calculated total Payload Mass carried by boosters.
- ✓ Counted successful landings at specific sites (e.g., CCAFS).
- ✓ Retrieved records of specific mission outcomes based on payload mass ranges.
- ✓ Ranked launch sites by successful landing counts using GROUP BY and ORDER BY.
- ✓ Performed time analysis to observe successful landing trends over specific months/years

GitHub URL for SQL Notebook:

https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA with SQL Results

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

EDA with SQL Results

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Total_Payload_Mass
```

```
45596
```

EDA with SQL Results

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Average_Payload_Mass
```

```
2928.4
```

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql SELECT MIN(Date) AS First_Successful_Landing_Date FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First_Successful_Landing_Date
```

```
2015-12-22
```

EDA with SQL Results

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) as Total_Number FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total_Number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA with SQL Results

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

EDA with SQL Results

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Month Landing_Outcome Booster_Version Launch_Site
```

```
01 Failure (drone ship) F9 v1.1 B1012 CCAFS LC-40
```

```
04 Failure (drone ship) F9 v1.1 B1015 CCAFS LC-40
```

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT Landing_Outcome, COUNT(*) as Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Landing_Outcome Outcome_Count
```

```
No attempt 10
```

```
Success (drone ship) 5
```

```
Failure (drone ship) 5
```

```
Success (ground pad) 3
```

```
Controlled (ocean) 3
```

```
Uncontrolled (ocean) 2
```

```
Failure (parachute) 2
```

```
Precluded (drone ship) 1
```

Folium Map:

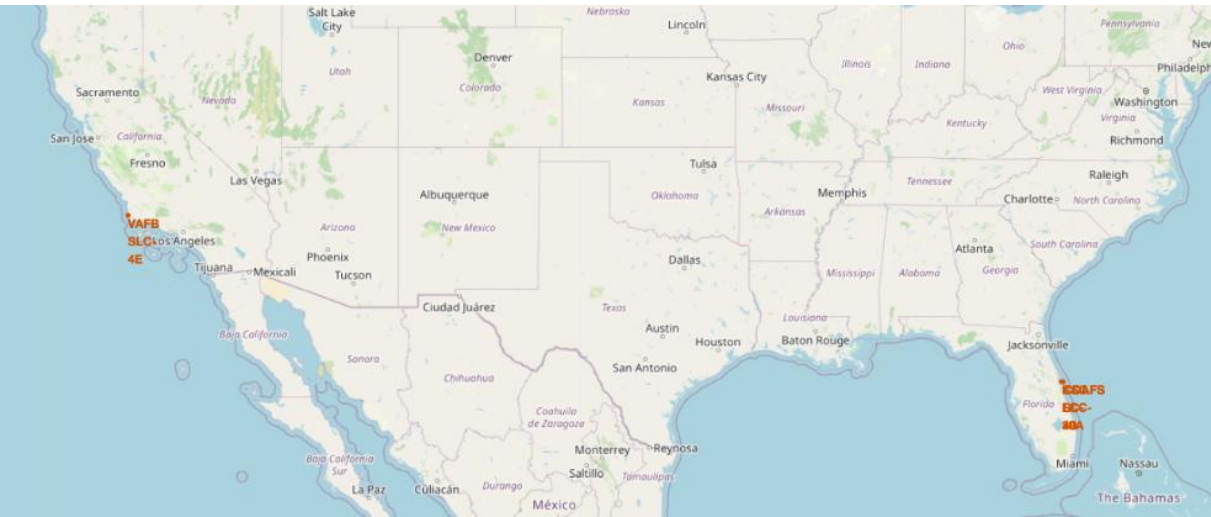
- Plotted Launch Sites on an interactive map.
- Added Marker Clusters to differentiate successful (green) and failed (red) launches.
- Performed proximity analysis by calculating distances to railways, highways, and coastlines.

Plotly Dash App:

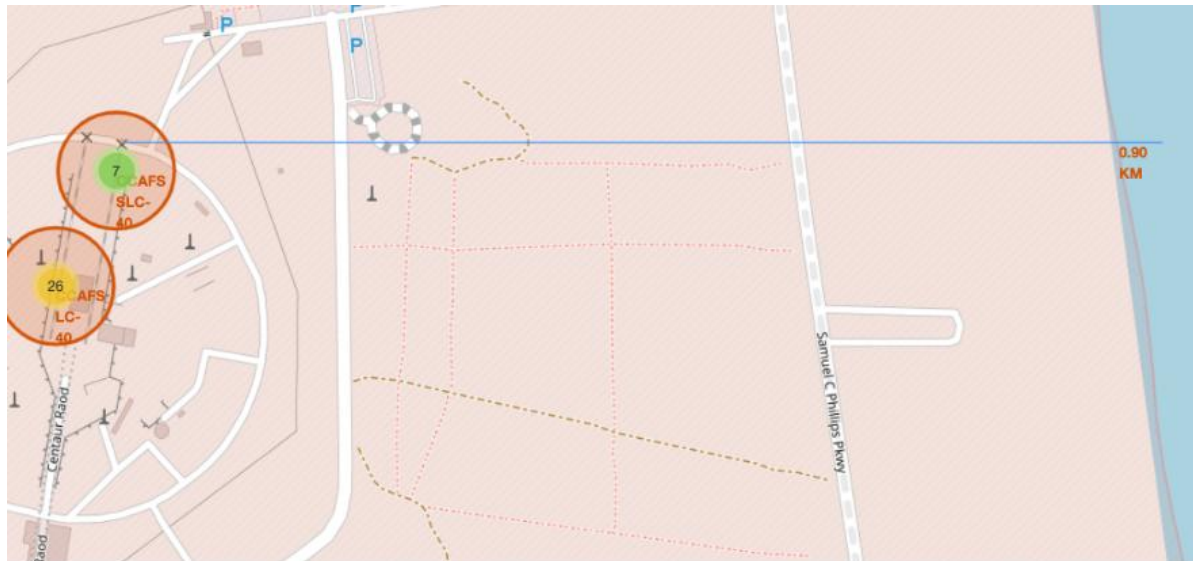
- Built a web dashboard with dropdowns and range sliders.
- Created interactive Pie Charts for success rates by site.
- Created Scatter Plots for Payload Mass vs. Outcome.

GitHub URLs: https://github.com/huyennt92/IBM-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Folium Map Results



Folium Map Results



Plotly Dash Results



Model Evaluation & Conclusion:

- **Models Tested:** Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- **Best Model:** All models performed similarly well on the test data (approx. 83.3% accuracy), but the **Decision Tree** (or SVM) is preferred for its specific hyperparameter tuning results.

Predictive Analysis

